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CS-322

Diamond Pricing

Introduction and Motivation

Diamond pricing is widely understood to depend on the "Four Cs": carat, cut, color, and clarity, a grading framework established by the Gemological Institute of America (GIA), which remains the international standard for evaluating diamond quality. Although these characteristics are commonly used by jewelers and consumers, their relative contribution to price is not always clear. Understanding which features most strongly influence diamond value can help inform purchasing decisions, guide industry pricing strategies, and improve transparency for buyers.

This project uses statistical modeling—specifically, multiple linear regression—to evaluate how the Four Cs and additional physical measurements (depth and table) affect diamond prices. The analysis draws on a dataset of 53,920 diamonds, providing a robust foundation for examining the drivers of price variation.

Data Description and Preparation

The dataset includes:

- Continuous variables: carat, depth, table, x, y, z, price
- Categorical variables: cut, color, clarity

Data Cleaning and Processing

- Invalid entries removed: Rows with zero values in x, y, or z were excluded.
- Log transformation: Because price distribution is highly skewed, *log(price)* was used as the response variable to stabilize variance.
- Ordinal encoding: Cut, color, and clarity follow natural industry rankings.
- Each was converted to an ordered numeric scale (e.g., Ideal = 5, Premium = 4), allowing regression.

This preprocessing ensures the model is both interpretable and statistically appropriate.

Data Overview

Carat: Weight of the diamond (1 carat = 0.2 grams)

- Carat determines the price; the larger the diamond is, the more it costs.

Cut: Quality of the diamond's cut,t which affects brilliance and sparkle.

- (worst to best (1 to 5): Fair, Good, Very Good, Premium, Ideal)
- Better cuts increase the visual appeal and price of the diamond.

Color: The color grade determines colorless to having a slight yellow tint.

- (best to worst (7 to 1): D, E, F, G, H, I, J)
- More colorless diamonds are rare and more expensive.

Clarity: Internal flaws and external marks on the diamond.

- (worst to best (1 to 8): I1 included in the end, SI2, SI1, VS2, VS1, VVS2, VVS1, IF)
- Diamonds with fewer inclusions are more valuable

Depth: Total depth percentage, how tall the diamond is relative to its width.

- Depth affects how light is refracted; extreme values can reduce brilliance and value.

Table: Width of diamond's top surface relative to the widest point as a percentage.

- Affects diamonds' sparkle and aesthetics.

X: Length of diamond in millimeters (mm).

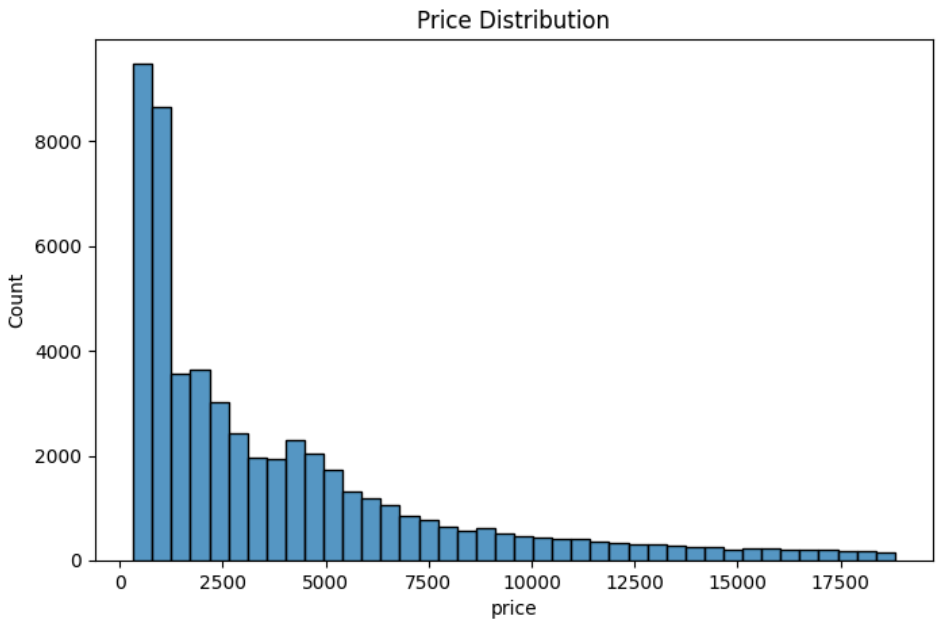
Y: Width of diamond in millimeters (mm).

Z: Depth of diamond in millimeters (mm) from top to bottom.

Price: Price of the diamond in US dollars ($).

Preliminary Calculation

Several visualizations were created to understand the dataset:



1. Price Distribution Histogram
Shows a right-skewed distribution, supporting the use of log transformation.

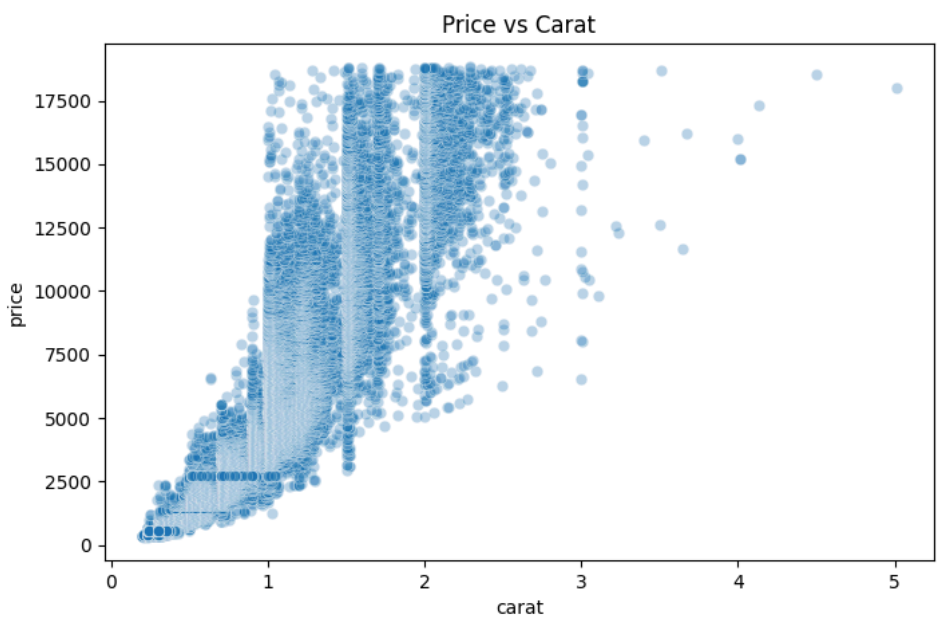


Figure 2. Price vs. Carat Scatterplot
Reveals a strong positive nonlinear relationship between carat weight and price.

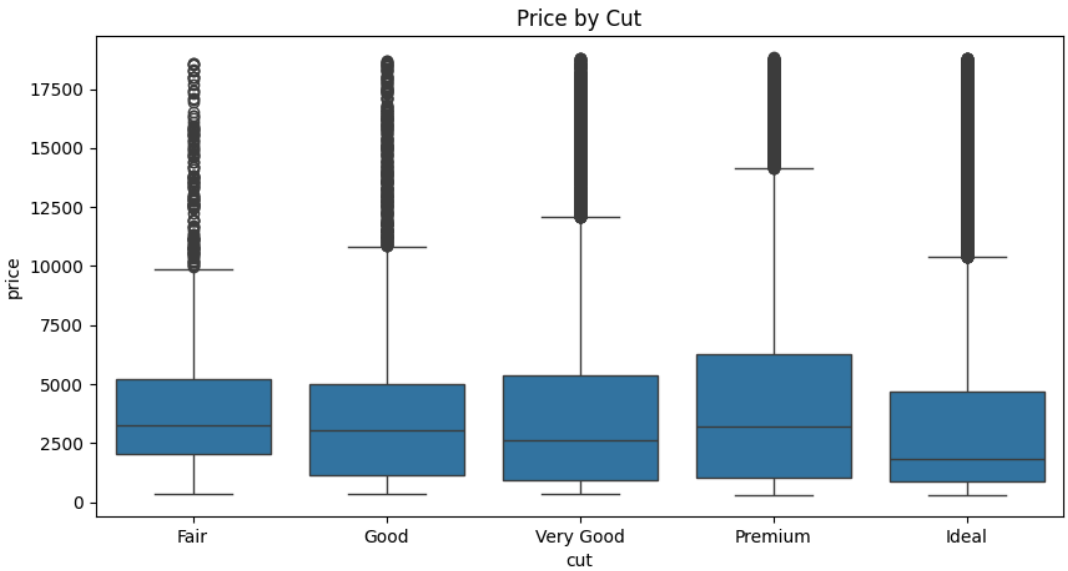


Figure 3. Price by Cut Boxplot

Indicates that higher-quality cuts (Ideal, Premium) tend to be associated with higher prices.

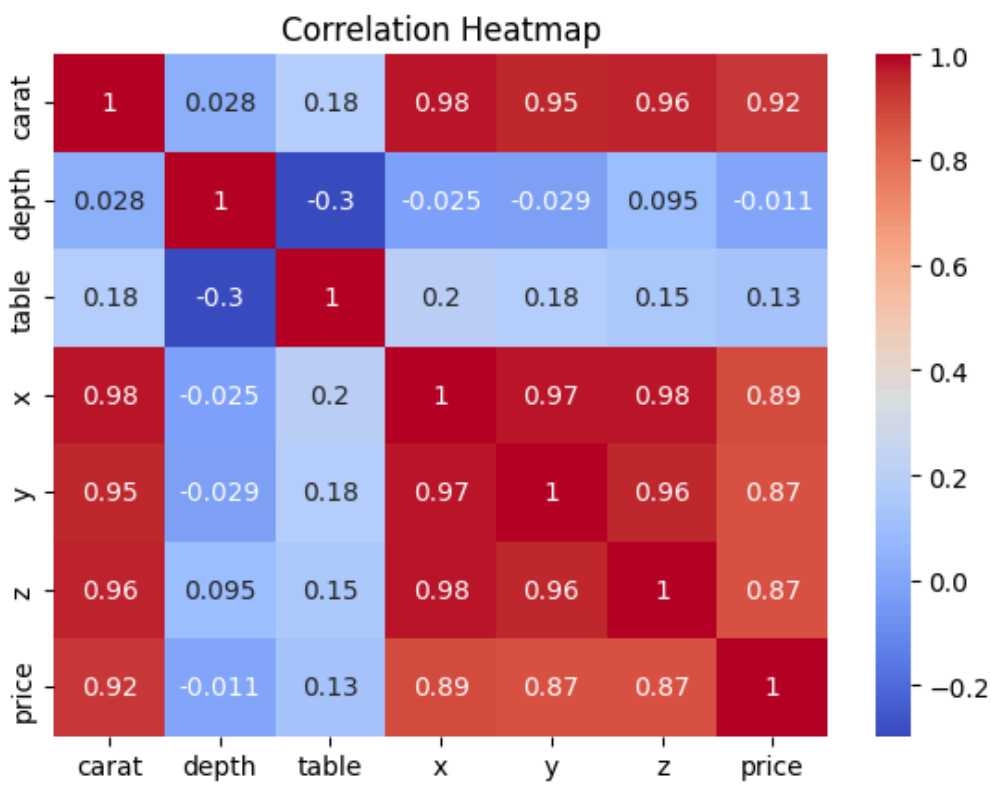


Figure 4. Correlation Heatmap

Highlights carat as the strongest single predictor and shows weak correlations for depth and table.

Regression Model

To understand how different diamond characteristics influence market price, a multiple linear regression model was fitted using the natural logarithm of price as the dependent variable. The log transformation was applied because raw diamond prices exhibit a highly right-skewed distribution. Using *log(price)* stabilizes the variance, improves normality of residuals, and allows coefficients to be interpreted as percentage changes in price rather than absolute dollar differences, which is more meaningful for exponential price growth patterns observed in gemstones.

The final model includes six predictors:

- **Carat** (continuous)
- **Depth** (continuous)
- **Table** (continuous)
- **Cut** – from Fair (1) to Ideal (5)
- **Color** – from J (1, lowest) to D (7, highest)
- **Clarity** – from I1 (1, lowest) to IF (8, highest)

The regression equation estimated is:

$$log(price) = \beta 0 + \beta 1(carat) + \beta 2(depth) + \beta 3(table) + \beta 4(cut) + \beta 5(color) + \beta 6(clarity)$$

Model Fit and Diagnostics

High explanatory power

- $R^2 = 0.880$
- Adjusted $R^2 = 0.880$

This means that 88% of the total variation in diamond prices is explained by the six predictors. In consumer pricing datasets, this level of explanatory power is unusually high, indicating that the chosen features are extremely strong determinants of price.

Model significance

- F-statistic = 66,020
- p-value < 0.001

The F-test confirms that the model significantly improves predictive accuracy compared to a model with no predictors.

Interpretation of Coefficients

Because the model uses *log(price)* as the dependent variable:

- Each coefficient represents the approximate percentage change in price for a one-unit increase in the predictor, holding all other variables constant.

Key findings

1. Carat shows the largest coefficient (2.1733), indicating a very strong exponential relationship.
    a. Every 1-unit increase in carat weight (e.g., from 0.5 to 1.5) is associated with a 217% increase in price, reflecting the well-known industry phenomenon that diamond prices rise faster than size increases.

2. Color and clarity also show large positive coefficients (around 0.086–0.090).
    a. Moving up one grade in color or clarity corresponds to an 8–9% increase in expected price. This confirms these factors are major drivers of diamond valuation beyond carat weight.

3. Cut has a smaller but significant effect (0.0204).
    a. Each improvement in cut grade (e.g., Good → Very Good) raises price by a modest 2%, highlighting that cut quality is important but not as influential as color, clarity, or carat weight.

4. Table size adds a small positive effect (0.0044).

5. Depth contributes a very small negative effect (-0.0024).
    a. These physical proportions have minimal economic influence compared to the Four Cs.

Regression Results

Predictor	Effect on *log(price)*	Significance (p-value)	Interpretation
Carat	**Strong positive effect**	**$p < 0.001$**	**Largest driver of price; heavier diamonds are much more expensive.**
Depth	**Negative effect**	**$p = 0.050$**	**No meaningful influence on price.**
Table	**Small positive effect**	**$p < 0.001$**	**Slight price increase with larger table measurements.**
Cut	**Small positive effect**	**$p < 0.001$**	**Better cut quality leads to slightly higher prices.**
Color	**Moderate positive effect**	**$p < 0.001$**	**Higher (less tinted) colors increase price.**
Clarity	**Moderate positive effect**	**$p < 0.001$**	**Higher clarity grades increase price.**

Summary:

- Carat is the dominant predictor.
- Color and clarity also strongly influence price.
- Cut and table have smaller effects.
- Depth is not practically relevant.

Hypothesis Testing

Predictor	Null Hypothesis (H_0)	Significance (p-value)	Decision	Interpretation	Alternative Hypothesis (H_1)
Carat	**Carat does not affect price**	**$p < 0.001$**	**Reject H_0**	**Strong positive effect**	**Carat does affect price**
Depth	**Depth does not affect price**	**$p = 0.050$**	**Fail to reject**	**No meaningful effect**	**Depth does affect price**
Table	**Table does not affect price**	**$p < 0.001$**	**Reject H_0**	**Small positive effect**	**Table does affect price**
Cut	**Cut does not affect price**	**$p < 0.001$**	**Reject H_0**	**Slight increase with better cut**	**Cut does affect price**
Color	**Color does not affect price**	**$p < 0.001$**	**Reject H_0**	**Higher color increases price**	**Color does affect price**
Clarity	**Clarity does not affect price**	**$p < 0.001$**	**Reject H_0**	**Higher clarity increases price**	**Clarity does affect price**

This study set out to investigate which physical and qualitative characteristics most strongly influence diamond prices using a comprehensive dataset of 53,920 diamonds. Through descriptive statistics, visualization, and multiple regression analysis, the findings reveal a clear and quantifiable structure underlying diamond valuation. The results support long-standing gemological principles while also providing precise statistical evidence for how each feature contributes to market price.

The analysis confirms that carat weight is by far the most influential factor in determining price. The strong positive coefficient on carat, combined with its extremely small p-value, demonstrates that diamond prices increase exponentially with size. This aligns with industry pricing models, where larger diamonds are exponentially rarer and therefore substantially more valuable.

Beyond carat weight, clarity and color emerge as the next strongest predictors, each contributing roughly 8–9% price increases per improvement in grade. This underscores that internal inclusions and body color meaningfully affect consumer perception and value. The study shows that these qualitative characteristics, while less influential than carat, still play a major role in determining overall price.

Cut quality, although statistically significant, showed a smaller economic effect. Better-cut diamonds do command higher prices, but the magnitude is modest compared to color, clarity, or carat. This suggests that while cut influences brilliance and appearance, buyers and sellers place more emphasis on rarity-driven characteristics when determining price.

Other structural attributes, such as depth and table percentages, were found to have minimal effects, indicating that dimension ratios are less relevant to pricing when quality grades are already accounted for.

Overall, the regression model explains 88% of the variation in diamond prices, demonstrating that diamond valuation is highly predictable when using the Four Cs and physical measurements. The hypothesis tests further validated that carat, cut, color, clarity, and table are all statistically significant contributors to price, with depth being only marginally significant.

This study highlights how statistical modeling can bring clarity and precision to processes traditionally guided by expert appraisal. By quantifying the economic value of gemological characteristics, the analysis reinforces the reliability of standardized grading systems used by consumers, jewelers, and certification agencies. Furthermore, the results offer practical insights for buyers and sellers seeking to understand how individual attributes contribute to market value.

In conclusion, the structured, data-driven approach used in this project provides strong evidence that diamond pricing is not arbitrary but instead follows a systematic pattern that can be accurately modeled. The findings confirm the central role of carat weight, the substantial influence of clarity and color, and the more modest but significant effect of cut. This comprehensive analysis contributes to a deeper understanding of gemstone economics and the factors shaping luxury market pricing.

References:

- **Gemological Institute of America (GIA). (n.d.).** ***Diamond 4Cs Guide.*** **https://www.gia.edu/diamond-4cs**

- **https://www.kaggle.com/datasets/zahranusrat/diamonds-characteristics-and-pricing-analysis?resource=download**